

Problem Set 2

Spring 2026

Due: 11:59 pm Monday, 23 Feb

To facilitate grading and timely feedback
please note that all submissions are through Gradescope.

Solve each problem on a new page.

Please clearly identify collaborator names and the resources used for each problem.

1. Argue whether each of the following statement is True or False. Give a short, high-level proof if the statement is True and a counter-example if the statement is False. Assume that $\Sigma = \{0, 1\}$ and that A and B are languages over Σ for each of the statements.

- (a) If A is nonregular, then there exists a nonregular language B such that $A \cap B$ is finite.
- (b) If A is nonregular, then there exists a nonregular language B such that $A \cap B$ is infinite.
- (c) If A is infinite, there must exist a language $B \subseteq A$ that is nonregular.
- (d) If A is nonregular, B is regular, and $A \cap B = \emptyset$, then $A \cup B$ is nonregular.

2. Let Σ be an alphabet. The symmetric difference of two languages $A, B \subseteq \Sigma^*$ is defined as

$$A \Delta B = (A \cap \overline{B}) \cup (\overline{A} \cap B).$$

- (a) Prove that if $A \subseteq \Sigma^*$ is a context-free language and $B \subseteq \Sigma^*$ is a finite language, then the language $A \Delta B$ is context-free.
 - (b) Give an example of an alphabet Σ , a context-free language $A \subseteq \Sigma^*$, an a regular language $B \subseteq \Sigma^*$ for which the language $A \Delta B$ is not context-free.
3. Give a context-free grammar for each of the following languages. You do not need to prove that your grammars are correct. However, if it is too difficult to verify the correctness of your grammars, you may lose points (even if they happen to be correct). You should therefore aim to give the simplest grammars possible, and it may be to your benefit to offer short explanations about how your grammars work.
 - (a) $L_1 = \{0^n 1^m \mid m \leq n \leq 2m\}$.
 - (b) $L_2 = \{w \in \{0, 1\}^* \mid w = w^R \text{ and the number of 1's in } w \text{ is divisible by 3}\}$.
 - (c) $L_3 = \{a^k b^n c^m \mid k, n, m \geq 0 \text{ and } k + n = m\}$.
 - (d) $L_4 = \{a^k b^n c^m \mid k, n, m \geq 0 \text{ and } k + m = n\}$.
 4. Let $C_{CFG} = \{\langle G, k \rangle \mid \text{Language of CFG } G \text{ contains exactly } k \text{ strings where } k \geq 0 \text{ or } k = \infty\}$. Show that C_{CFG} is decidable.
 5. Sheikh Chilly has bought the latest Turing Machine model TM202x for mining cryptocurrency that uses one-sided unbounded (on the left) tape. Unfortunately the TM was damaged during delivery such that that transition function can only implement the following mapping.

$$\delta : Q \times \Gamma \mapsto Q \times \Gamma \times \{L, BACK\}.$$

If $\delta(q, a) = (r, b, BACK)$, when the machine is in state q reading an a , the machine's head jumps back to the right-hand end of the tape after it writes b on the tape and enters state r . In other words, instead of being able to move the head one cell right, it can only move back to the start of the tape on the right end. Show Sheikh Chilly how he can program this TM202x to simulate an undamaged TM.

6. *Bonus Challenge Problem:* What is the smallest NFA you can design for $\{a^n : n \neq 1003\}$. You should argue why your NFA is correct.